



- 6 Nine athletes in a club have a new coach. The coach adopts a new training programme which he believes will reduce the race times of these athletes. Each athlete completes a 1500 m time trial before and after completing the new training programme. Their times, in seconds (s), are recorded.

Athlete	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>
Time before training (s)	250	251	252	267	276	291	310	320	335
Time after training (s)	245	251	253	261	275	293	302	313	320

- (a) Carry out a paired t -test at the 5% significance level to test the coach's belief. [7]

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Further research suggests that the effects of the training programme tend to reduce the times of the slower athletes by more than those of the faster athletes.

- (b) Suggest a reason why the paired t -test used in part (a) may not have been an appropriate test in this case. [1]

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- (c) Suggest a suitable alternative test that could have been used instead of a paired t -test. [1]

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